

Enforcing Neurorights: A Practical Framework for Preventing Covert Brain-Data Collection in Homes and Virtual Worlds

The rapid advancement of neurotechnologies capable of capturing and analyzing brain impulses presents a profound challenge to established legal and ethical frameworks ¹⁶. These technologies, ranging from consumer-grade EEG headsets to sophisticated Brain-Computer Interfaces (BCIs), blur the line between mental activity and data, creating urgent risks for mental privacy ⁵⁴. This report outlines a comprehensive research framework designed to defend neurorights by establishing a dual-layered system of preventive legal checks and post-hoc analysis tools. The focus is on two critical contexts where consent must be treated with utmost care: private residences and virtual ecosystems such as VR/AR platforms. Grounded primarily in the emerging legal landscape of U.S. state and federal neural-data laws, the framework incorporates comparative insights from the European Union's AI Act and General Data Protection Regulation (GDPR). It is anchored in the foundational doctrine of international human rights, specifically the rights to mental privacy, freedom of thought, and cognitive liberty, providing a robust intellectual and legal foundation for action.

The Foundational Doctrine: Human Rights, Mental Privacy, and Cognitive Liberty

The defense of neurorights requires a firm grounding in a coherent and compelling doctrinal framework that moves beyond conventional data protection. The central premise is that neural data is not merely another category of personal information; it is a direct and intimate proxy for an individual's thoughts, intentions, emotional states, and sense of self ^{11 54}. This unique characteristic elevates the need for protection from general privacy principles to a distinct set of rights focused on the sanctity of the mind. The discourse has coalesced around three interconnected concepts: mental privacy,

mental integrity, and cognitive liberty [17](#) . Mental privacy refers to the right to keep one's mental life free from unauthorized intrusion, observation, or access [17](#) [69](#) . This concept is often considered a component of the broader right to privacy enshrined in instruments like Article 8 of the European Convention on Human Rights (ECHR), which has been interpreted to cover activities such as taking photographs on arrest or retaining fingerprints [37](#) [103](#) . However, the advent of neurotechnology tests the limits of this interpretation, as it potentially allows for a form of "brain-reading" that goes far beyond traditional surveillance [103](#) .

Mental integrity is the corollary right to protect one's mind from harmful external interference, manipulation, or alteration [36](#) [48](#) . This directly addresses the threat posed by technologies that could be used to coerce, influence, or reshape an individual's beliefs or behaviors without their genuine consent. Proposed neurorights include a "right to mental agency," or free will, and a "right to mental identity," or a sense of self, which are both threatened by such interventions [20](#) . Finally, cognitive liberty encompasses the freedom to control one's own cognitive processes, including the ability to choose what information to process, how to think, and whether to voluntarily engage with neurotechnologies that augment or alter cognition [17](#) [65](#) . UNESCO's Recommendation on the Ethics of Artificial Intelligence emphasizes that neurotechnology should empower individuals to make free and informed decisions about their nervous system and mental health, reflecting the core tenets of cognitive liberty [28](#) .

These rights are deeply rooted in international human rights law, which accords the freedom of thought and conscience a high degree of protection, often described as almost absolute [97](#) [98](#) . While existing human rights law may be considered an "incomplete" framework for these new challenges, its flexibility allows for development through jurisprudence [104](#) . The United Nations has explicitly called for the regulation of neurotechnologies to protect the right to mental privacy [87](#) . The most significant recent development in this area is the landmark August 2023 ruling by the Chilean Supreme Court against Emotiv, a US-based neurotechnology company [64](#) [71](#) . In this case, the court established that brain activity is constitutionally protected, effectively treating it as a special category of data deserving of heightened safeguards. This precedent demonstrates that constitutional-level protection for mental privacy is not only theoretically possible but practically achievable, providing a powerful model for policymakers globally. By anchoring the defense of neurorights in this robust human rights doctrine, advocates and regulators can frame the issue not just as a matter of data security, but as a fundamental struggle to preserve the last bastion of individual autonomy in an increasingly surveilled world [55](#) .

The Primary Legal Landscape: U.S. State and Federal Neural Data Regulations

While international human rights provide the philosophical foundation, the immediate and actionable framework for defending neurorights in the United States is emerging from a dynamic and accelerating wave of state-level legislation. This trend creates a fragmented but increasingly protective patchwork of laws that businesses must navigate. The pioneers in this movement are Colorado and California, which have amended their comprehensive privacy acts to explicitly define neural data as "sensitive" data [45](#) [46](#) [47](#) . Under frameworks analogous to the GDPR, classifying data as sensitive triggers significantly stricter rules regarding its collection, use, and sharing. For instance, under GDPR guidelines, valid consent for sensitive data must be freely given, specific, informed, and unambiguous, and it must involve a clear affirmative act by the data subject [76](#) [77](#) . This means that any entity collecting neural data in Colorado or California must secure explicit, purpose-limited consent and cannot easily repurpose the data for unrelated commercial activities like advertising or AI training without a separate, valid legal basis [9](#) .

This legislative surge is not confined to these two states. At least 15 additional bills targeting mental privacy and neural data are pending in various state legislatures across the country, with more anticipated [71](#) [91](#) . These proposed laws consistently emphasize core principles such as transparency, explicit consent, and strict limitations on data use, indicating a broad consensus among lawmakers that general-purpose privacy laws are insufficient for the unique threats posed by neurotechnology [71](#) . This wave of state activity highlights the need for a flexible compliance framework that can adapt to jurisdiction-specific nuances. For example, while Colorado and California treat neural data as sensitive, other states may adopt different classifications or create entirely new regulatory categories [45](#) .

Complementing the state-by-state approach, there is growing awareness at the federal level. The proposed Management of Individuals' Neural Data (MIND) Act represents a significant step toward creating a uniform national standard [25](#) . If enacted, the MIND Act would define neural data and empower the Federal Trade Commission (FTC) to regulate its collection, sharing, and use, thereby preempting conflicting state laws and providing clearer guidance for industry [25](#) [72](#) . Although still in the proposal stage, the existence of the MIND Act signals a maturing debate and provides a valuable template for what a future federal framework might look like. Together, the robust protections emerging from states like Colorado and California, the broader legislative momentum across the nation,

and the aspirational goals of federal proposals like the MIND Act form the primary legal architecture upon which a defense strategy for neurorights can be built.

Jurisdiction	Key Legislation / Proposal	Treatment of Neural Data	Core Requirements
United States	MIND Act (Proposed)	Defines neural data and empowers the FTC to regulate its collection, sharing, and use. 25 72	Not yet law; would establish a federal standard.
Colorado	Colorado Privacy Act (amended)	Classified as "sensitive" data. 45 46	Requires explicit, specific, revocable consent and strict purpose limitation. 76
California	California Consumer Privacy Act (amended)	Classified as "sensitive" data. 46 47	Requires explicit, specific, revocable consent and strict purpose limitation. 76
Montana & Connecticut	State Privacy Laws (amended)	Classified as "sensitive" data. 45	Imposes stricter rules for consent and use similar to GDPR principles. 116
Multiple States	Pending Mental-Privacy Bills	Explicitly targets neural data collection and use. 71 91	Mandates transparency, explicit consent, and use restrictions.

Comparative Benchmarks: EU AI Act and GDPR as Gold Standards

While U.S. law forms the primary operational framework, the European Union's regulatory approach serves as a critical benchmark for best practices and aspirational goals. The EU's framework is notable for its proactive, risk-based approach and its explicit, enforceable prohibitions on certain types of neurotechnology applications. The cornerstone of this approach is the EU AI Act, which introduces a ban on AI systems that use "significantly harmful subliminal manipulation" [93](#) [94](#). This provision, found in Article 5(1)(a), is a direct and powerful tool for challenging covert neuro-nudging in environments like virtual reality platforms [121](#). If a VR application uses real-time neural feedback to infer a user's attention or emotional state and then deploys subtle, non-obvious cues to manipulate their behavior—for example, guiding them to make a purchase—it would likely fall under this prohibition [66](#). The act sets a high bar by focusing on practices that "significantly distort" an individual's behavior, moving beyond mere deception to address coercive interference with cognitive autonomy [106](#).

Another crucial element of the EU AI Act is its prohibition on the use of emotion recognition AI in specific high-stakes contexts. The act bans the use of such technology for workplace employee monitoring and in educational institutions for grading or

recruitment purposes, except for narrow exceptions related to safety or medical treatment . This provides a clear and unambiguous rule for preventing the misuse of inferred emotional and cognitive states in professional and academic settings. This aligns closely with concerns driving similar legislation in the U.S., where lawmakers are wary of employers using neural data for performance evaluations or schools using it for disciplinary actions [71](#) . By adopting such explicit bans, the EU establishes a strong normative position that prioritizes human dignity and autonomy over the potential efficiency gains offered by pervasive affective computing.

Beyond the AI Act, the General Data Protection Regulation (GDPR) already provides a high level of protection for many forms of neurodata. Under GDPR, certain types of neurodata are classified as "special category data," placing them in the same protected tier as genetic data and data concerning health [14](#) [116](#) . Processing such data requires a strong legal basis, with explicit consent being one of the most common requirements [116](#) . Furthermore, the GDPR definition of biometric data for the purpose of uniquely identifying a person includes a requirement for "functional use," which reinforces the principle of purpose limitation and makes it harder to repurpose data collected for one function for another [12](#) . The combination of the AI Act's targeted prohibitions and the GDPR's stringent baseline protections for sensitive data offers a comprehensive and robust model. Advocates and regulators in the U.S. can leverage this framework as a "gold standard" to argue for stronger, more explicit protections than those currently available under general privacy laws, pushing for regulations that not only govern data handling but also proscribe inherently risky applications of neurotechnology.

Contextual Application I: Preventing Unlawful Practices in Private Residences

The home has traditionally been considered the last bastion of privacy, a sanctuary from the prying eyes of the market and the state [55](#) . The introduction of neurotechnologies into this space—such as headsets for meditation, sleep tracking, or even entertainment—fundamentally alters this equation. The primary risk profile in private residences revolves around the erosion of this sanctity through covert, overly broad, or deceptive data collection practices [55](#) . Even when collection occurs with a user's consent, the lack of transparency regarding how that data is used, shared, or stored remains a major concern. A device marketed for therapeutic focus enhancement could easily begin collecting granular data on a user's cognitive load and stress levels, which is then sold to third

parties for behavioral advertising or used to train proprietary AI models without the user's full understanding or meaningful consent ⁹ .

In this context, the legal framework demands exceptionally robust safeguards. Consent must be more than a pre-checked box in a lengthy terms-of-service agreement; it must be explicit, specific, granular, revocable, and strictly purpose-limited ⁹ . This means a user must give clear permission for each distinct use of their neural data, and they must have an equally clear and easy way to withdraw that permission at any time without penalty ⁷⁶ . The classification of neural data as "sensitive" under the laws of Colorado and California imposes these requirements automatically, making any sale or secondary use of data for unrelated commercial purposes highly suspect and likely unlawful ^{45 47} . Companies operating in these jurisdictions must design their systems to respect these boundaries from the outset. For example, if a BCI headset collects neural data during a workout, it cannot assume the right to use that data to infer mood patterns and sell targeted wellness product recommendations unless the user has given explicit, separate consent for that specific purpose.

A successful defense of neurorights in the private residence hinges on preventive measures. Internal corporate compliance protocols must be developed to embed these principles into the product lifecycle. This involves creating detailed checklists that product teams, engineers, and privacy officers must follow before launching any feature involving neural data. These checklists should mandate transparency in user interfaces, ensuring that data collection is clearly disclosed in plain language. They should also verify that the consent mechanism is truly affirmative and granular, allowing users to opt-in to specific functions rather than accepting a blanket policy. Furthermore, these protocols must include technical and procedural safeguards to prevent data from being repurposed or aggregated in ways that could re-identify individuals or build detailed psychological profiles without a valid legal basis ¹³ . By shifting the focus to design-time constraints, companies can avoid the more severe penalties and reputational damage associated with post-deployment violations, while also respecting the fundamental right of individuals to maintain the privacy of their innermost thoughts within their own homes.

Contextual Application II: Defending Neurorights in Virtual Ecosystems

Virtual and augmented reality ecosystems present a new frontier for neurorights challenges, characterized by complex interactions between immersive environments and real-time neural feedback. Unlike the static nature of data collection in a private residence, the risks in VR/AR are dynamic and often tied to user engagement within a platform. The primary danger lies in the potential for covert influence and behavioral manipulation. Advanced BCIs integrated with VR environments could detect a user's intention to interact with an object, their level of attention, or their emotional state ²⁹ ³¹. This data, processed by machine learning algorithms, could be used to deliver "subliminal nudges"—micro-adjustments to the environment, content, or pricing designed to guide a user's choices without their conscious awareness ⁶⁶. This practice directly contravenes the spirit, if not the letter, of emerging regulations. The EU AI Act's explicit ban on AI systems using subliminal techniques to significantly distort behavior provides a clear legal basis for challenging such manipulative designs ⁹³ ¹²¹.

Another significant risk in virtual ecosystems is large-scale behavioral profiling, particularly concerning vulnerable populations like children. An AR game, for instance, could infer a child's frustration or boredom levels and dynamically adjust gameplay difficulty or trigger targeted advertisements for relaxation apps or energy drinks ⁶⁶. This raises serious ethical questions about exploitation and the commercialization of a child's cognitive and emotional state. The wave of state legislation in the U.S. targeting mental privacy is partly motivated by such scenarios, aiming to create a safer digital environment for minors ⁹¹. Moreover, the integration of BCIs with the metaverse blurs the lines between the user and their avatar, raising profound questions about the authenticity of experience and the preservation of cognitive liberty ³⁰ ⁶⁵. When a user's thoughts can directly control elements within a virtual world, the boundary between self and system dissolves, creating novel challenges for personal identity and agency.

To defend neurorights in these complex environments, the legal and technical frameworks must evolve accordingly. Platform developers must conduct rigorous ethical reviews, such as the Operational Neuroethical Risk Assessment and Mitigation Protocol for Artificial Intelligence in Virtual Reality (ONRAMP-AI-VRAR), to identify and mitigate potential harms before deployment ¹. Compliance checklists for VR/AR companies should specifically flag practices like emotion inference for micro-targeting, especially in applications aimed at children. For advocacy groups and regulators, the EU AI Act's prohibitions serve as powerful diagnostic tools. When a new VR/AR application launches,

investigators can use these benchmarks to assess whether the platform's use of neural data constitutes prohibited manipulation or unfair profiling. This requires a deep understanding of both the technology and the law, enabling stakeholders to hold platforms accountable for protecting users' mental integrity and cognitive freedom within these increasingly influential digital spaces.

Operationalizing Defense: From Research to Actionable Workflows

The ultimate goal of this research is to translate legal doctrine and technological understanding into concrete, actionable workflows that can be deployed by companies, advocates, and regulators. This operationalization creates a resilient defense system capable of both preventing harm and remedying it when it occurs. The framework is built on a symbiotic relationship between law, technology, and advocacy. Law provides the essential blueprint of rights and obligations, technology acts as a translator that makes this law dynamic and applicable in real-world contexts, and advocacy serves as the enforcement mechanism that drives accountability. This synergy creates a virtuous cycle: legal requirements push companies toward safer designs; advocacy exposes non-compliance; and regulatory action reinforces the legal requirements, leading to better industry-wide practices.

The first layer of this operational framework focuses on prevention through **internal human-in-the-loop review protocols for companies**. Before any product or feature involving neural data is launched, it must pass through a rigorous compliance checklist grounded in the synthesized legal findings. This checklist, derived from U.S. state laws, the EU AI Act, and human rights principles, would force product teams to confront key ethical and legal questions upfront. For example, it would require verification that there is written, purpose-specific consent for every data processing activity ⁷⁶. It would demand the presence of a clear opt-out mechanism that does not unfairly penalize users who choose to disengage ⁹. It would also scrutinize whether data is being repurposed for unrelated functions like advertising or AI training, which would likely violate the principle of purpose limitation ⁴⁵. By embedding these checks into the development lifecycle, companies can proactively avoid violations and build products that respect user rights by design.

The second layer empowers **advocacy and oversight workflows for civil society**. Non-governmental organizations, journalists, and community watchdogs need standardized tools to evaluate the compliance of neurotech products and platforms. These toolkits would provide a step-by-step methodology for assessing a product's policies and practices against the established neurorights framework. The workflow would involve identifying the technology in question, classifying its functions according to a predefined schema (e.g., `virtual_platform_tracking`), querying a centralized database of relevant laws (`neural_policies`), and generating a compliance report that highlights potential violations ¹. This empowers watchdogs to move beyond anecdotal criticism and produce structured, evidence-based assessments that can inform public debate and pressure companies to change their practices.

The final layer consists of **regulatory submission templates** for formal complaints and petitions. When advocacy efforts reveal clear violations, these templates provide a ready-made format for filing official complaints with relevant authorities, such as the FTC, state Attorneys General, or the European Commission. Each template would be pre-populated with sections mapping the specific facts of an incident to the precise legal provisions that were likely breached. For instance, a complaint against a VR platform engaging in subliminal manipulation could directly cite Article 5(1)(a) of the EU AI Act, while a complaint about a company selling neural data for unrelated marketing purposes in Colorado would reference the relevant amendments to the Colorado Privacy Act ^{45 93}. By providing these standardized tools, the framework lowers the barrier to entry for regulatory engagement, enabling a wider range of stakeholders to participate in the enforcement of neurorights and helping to close the gap between emerging legal protections and their real-world implementation.

Reference

1. ONRAMP-AI-VRAR: an operational protocol for ethics and ... - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC12696944/>
2. (PDF) The use of CNNs in VR/AR/MR/XR: a systematic literature ... https://www.researchgate.net/publication/383566072_The_use_of_CNNs_in_VRARMRXR_a_systematic_literature_review
3. The Role of Machine Learning in AR/VR-Based Cognitive Therapies <https://www.mdpi.com/2079-9292/14/6/1110>

4. TLD: Deep Learning-Based Automated Privacy Policy Annotation ... <https://dl.acm.org/doi/10.1145/3463676.3485608>
5. [PDF] Intelligent World 2030 - Huawei https://www-file.huawei.com/-/media/corp2020/pdf/giv/intelligent_world_2030_en.pdf
6. Research on the cognitive neural mechanism of privacy ... - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC11018622/>
7. Rethinking Data Protection in the (Generative) Artificial Intelligence ... <https://arxiv.org/html/2507.03034v1>
8. Unraveling the Neurocognitive Mechanisms of Delayed Punishment ... <https://www.sciencedirect.com/science/article/pii/S1053811925005774>
9. (PDF) The Ethics of Artificial Intelligence: An Introduction https://www.researchgate.net/publication/365023130_The_Ethics_of_Artificial_Intelligence_An_Introduction
10. EACS Abstracts - 2023 - HIV Medicine - Wiley Online Library <https://onlinelibrary.wiley.com/doi/full/10.1111/hiv.13572>
11. Beyond neural data: Cognitive biometrics and mental privacy <https://www.sciencedirect.com/science/article/pii/S0896627324006524>
12. [PDF] EUROPEAN COMMISSION Brussels, 29.7.2025 C(2025) 5052 final ... https://ai-act-service-desk.ec.europa.eu/sites/default/files/2025-08/guidelines_on_prohibited_artificial_intelligence_practices_established_by_regulation_eu_20241689_ai_act_english_ied3r5nwo50xggpcfmwckm3nuc_112367-1.PDF
13. The EU GDPR and secondary use of health and genetic data for ... <https://academic.oup.com/idpl/advance-article/doi/10.1093/idpl/ipag001/8443007>
14. (PDF) Neurodata and Neuroprivacy: Data Protection Outdated? https://www.researchgate.net/publication/265048889_Neurodata_and_Neuroprivacy_Data_Protection_Outdated
15. The Marriage of Neurotechnologies and Artificial Intelligence - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC12965113/>
16. Protecting Brain Privacy in the Age of Neurotechnology https://www.researchgate.net/publication/384971350_Protecting_Brain_Privacy_in_the_Age_of_Neurotechnology_Policy_Responses_and_Remaining_Challenges
17. [PDF] Protecting Cognition: Human Rights and Neurotechnology - ohchr <https://www.ohchr.org/sites/default/files/documents/hrbodies/hrcouncil/advisorycommittee/neurotechnology/02-nhris/ac-submission-nhri-australia.pdf>
18. [PDF] A/HRC/58/58/Add.2 - General Assembly - the United Nations <https://docs.un.org/en/A/HRC/58/58/Add.2>

19. [PDF] International Human Rights Protection Gaps in the Age of ... <https://ntc.columbia.edu/wp-content/uploads/2022/05/NeurorightsFoundationPUBLICAnalysis5.6.22.pdf>
20. [PDF] Neurotechnology - ohchr <https://www.ohchr.org/sites/default/files/documents/hrbodies/hrcouncil/advisorycommittee/neurotechnology/03-ngos/ac-submission-cso-neurorightsfoundation.pdf>
21. Investigating neural impairments in psychotic disorders using ... - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC12949052/>
22. Human-specific features of the cerebellum and ZP2-regulated ... <https://www.sciencedirect.com/science/article/pii/S0092867426002217>
23. Multimodal dataset linking wide - field calcium imaging to behavior ... <https://www.nature.com/articles/s41597-025-05482-y>
24. The Temple University Hospital EEG Data Corpus - Frontiers <https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2016.00196/full>
25. Asymmetric depth - filtration: A versatile and scalable method for high ... <https://isevjournals.onlinelibrary.wiley.com/doi/10.1002/jev2.12256>
26. Poster Presentations - 2019 - Haemophilia - Wiley Online Library <https://onlinelibrary.wiley.com/doi/10.1111/hae.13666>
27. An open-source human-in-the-loop BCI research framework - Frontiers <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2023.1129362/full>
28. Draft text of the Recommendation on the Ethics of Neurotechnology <https://unesdoc.unesco.org/ark:/48223/pf0000393395>
29. Rewiring Attention: Virtual Reality and Brain-Computer Interfaces in ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC12898536/>
30. Meta-Review on Brain-Computer Interface (BCI) in the Metaverse <https://dl.acm.org/doi/full/10.1145/3696109>
31. [PDF] Non-Invasive Brain Computer Interfaces - HAL <https://hal.science/hal-04924929v1/file/BCI%20CHAPTER%20KARWOWSKI%20FINAL%20DRAFT%20%204%2019%202023%20FIN%20wk%20%281%29.pdf>
32. [PDF] Action research on the impact of neurotechnology on lifelong ... https://www.itcilo.org/sites/default/files/2025-09/Action%20research%20on%20the%20impact%20of%20neurotechnology%20on%20lifelong%20learning_WEBrev2.pdf
33. When Brain-Computer Interfaces meet the metaverse <https://www.sciencedirect.com/science/article/pii/S0925231225002097>

34. Forensic Brain-Reading and Mental Privacy in European Human ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC7612400/>
35. [PDF] The protection of mental privacy in the area of neuroscience [https://www.europarl.europa.eu/RegData/etudes/STUD/2024/757807/EPRS_STU\(2024\)757807_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/STUD/2024/757807/EPRS_STU(2024)757807_EN.pdf)
36. right to mental integrity in the age of neurotechnology: constructing ... <https://academic.oup.com/jlb/article/12/1/laf010/8156051>
37. The Right to Mental Privacy (Chapter 4) - Minds, Freedoms and Rights <https://www.cambridge.org/core/books/minds-freedoms-and-rights/right-to-mental-privacy/2ACCA293F89826C2902AD701F267241D>
38. [PDF] Tactful Inattention: Erving Goffman, Privacy in the Digital Age, and ... <https://core.ac.uk/download/pdf/217002391.pdf>
39. Sensors, Volume 24, Issue 16 (August-2 2024) – 371 articles - MDPI <https://www.mdpi.com/1424-8220/24/16>
40. Augmented Cognition - Springer Nature <https://link.springer.com/content/pdf/10.1007/978-3-031-93724-8.pdf>
41. Latest articles - MDPI https://www.mdpi.com/latest_articles
42. Sensors, Volume 25, Issue 7 (April-1 2025) – 389 articles <https://www.mdpi.com/1424-8220/25/7>
43. Sensors, Volume 23, Issue 2 (January-2 2023) – 492 articles <https://www.mdpi.com/1424-8220/23/2>
44. Appl. Sci., Volume 12, Issue 3 (February-1 2022) – 830 articles <https://www.mdpi.com/2076-3417/12/3>
45. [PDF] Comparing New Neural Data Privacy Laws In 4 States <https://cdp.cooley.com/wp-content/uploads/2025/06/Law360-Comparing-New-Neural-Data-Privacy-Laws-In-4-States.pdf>
46. States Pass Privacy Laws Safeguarding Brain Data Collected <https://www.govtech.com/policy/states-pass-privacy-laws-safeguarding-brain-data-collected>
47. California Passes Law to Protect Consumer 'Brain Data' - GovTech <https://www.govtech.com/health/california-passes-law-to-protect-consumer-brain-data>
48. The right to mental integrity in the age of neurotechnology - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC12133093/>
49. (PDF) Minding Rights: Mapping Ethical and Legal Foundations of ... https://www.researchgate.net/publication/370771193_Minding_Rights_Mapping_Ethical_and_Legal_Foundations_of_'Neurorights'

50. Minding Rights: Mapping Ethical and Legal Foundations of ... <https://www.cambridge.org/core/journals/cambridge-quarterly-of-healthcare-ethics/article/minding-rights-mapping-ethical-and-legal-foundations-of-neurorights/2F3BD282956047E1E67AA9049A2A0B68>
51. [PDF] shaping a rights - oriented digital transformation | oecd https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/06/shaping-a-rights-oriented-digital-transformation_30378a18/86ee84e2-en.pdf
52. Recommendation on the Ethics of Artificial Intelligence - UNESCO <https://www.unesco.org/en/articles/recommendation-ethics-artificial-intelligence>
53. Recommendation on the ethics of artificial intelligence https://digitallibrary.un.org/record/4062376?ln=zh_CN
54. Mental privacy: navigating risks, rights and regulation - PMC - NIH <https://pmc.ncbi.nlm.nih.gov/articles/PMC12287510/>
55. Your Brain Is the Last Private Space. Neurotech Companies Want In. <https://dev.to/tiamatenity/your-brain-is-the-last-private-space-neurotech-companies-want-in-447d>
56. HCI in Mobility, Transport, and Automotive Systems - Springer Nature <https://link.springer.com/content/pdf/10.1007/978-3-031-35678-0.pdf>
57. [XLS] queryResult - FDA <https://www.fda.gov/media/186045/download>
58. Grifols, SA - SEC.gov https://www.sec.gov/Archives/edgar/data/1438569/000110465925017501/tm257688d1_6k.htm
59. J. Clin. Med., Volume 14, Issue 23 (December-1 2025) – 354 articles <https://www.mdpi.com/2077-0383/14/23>
60. (PDF) Towards new human rights in the age of neuroscience and ... https://www.researchgate.net/publication/316473442_Towards_new_human_rights_in_the_age_of_neuroscience_and_neurotechnology
61. from brain evidence to neurorights: rethinking criminal responsibility ... https://www.academia.edu/144397239/FROM_BRAIN_EVIDENCE_TO_NEURORIGHTS_RETHINKING_CRIMINAL_RESPONSIBILITY_IN_THE_AGE_OF_NEUROTECHNOLOGY
62. human rights systems of protection from neurotechnologies that alter ... https://www.researchgate.net/publication/370596028_HUMAN_RIGHTS_SYSTEMS_OF_PROTECTION_FROM_NEUROTECHNOLOGIES_THAT_ALTER_BRAIN_ACTIVITY
63. (PDF) Persuasive technologies and the right to mental liberty https://www.researchgate.net/publication/344769042_Persuasive_technologies_and_the_right_to_mental_liberty_The_'smart'_rehabilitation_of_criminal_offenders

64. (PDF) Chilean Supreme Court ruling on the protection of brain activity https://www.researchgate.net/publication/378518025_Chilean_Supreme_Court_ruling_on_the_protection_of_brain_activity_neurorights_personal_data_protection_and_neurodata
65. (PDF) Neuralink's Brain-Computer Interfaces and the Reshaping of ... https://www.researchgate.net/publication/393189625_Neuralink's_Brain-Computer_Interfaces_and_the_Reshaping_of_Religious-Psychological_Experience
66. Understanding the Ethical Issues of Brain-Computer Interfaces (BCIs) https://www.researchgate.net/publication/379827163_Understanding_the_Ethical_Issues_of_Brain-Computer_Interfaces_BCs_A_Blessing_or_the_Beginning_of_a_Dystopian_Future
67. First report on the implementation of the 2021 Recommendation on ... <https://unesdoc.unesco.org/ark:/48223/pf0000391341>
68. Recommendation on the Ethics of Artificial Intelligence - UNESCO <https://www.unesco.org/en/articles/recommendation-ethics-artificial-intelligence-0>
69. Neurorights, Mental Privacy, and Mind Reading - Springer Nature <https://link.springer.com/article/10.1007/s12152-024-09568-z>
70. Forensic Brain-Reading and Mental Privacy in European Human ... https://www.researchgate.net/publication/342337728_Forensic_Brain-Reading_and_Mental_Privacy_in_European_Human_Rights_Law_Foundations_and_Challenges
71. Unlocking Neural Privacy: The Legal and Ethical Frontiers ... - Cooley <https://www.cooley.com/news/insight/2025/2025-03-13-unlocking-neural-privacy-the-legal-and-ethical-frontiers-of-neural-data>
72. Senators call for probe into neural data collection and security <https://www.fiercebiotech.com/medtech/senators-call-probe-neural-data-collection-and-security-practices>
73. Neuroscience Meets Cybersecurity - Springer Nature <https://link.springer.com/content/pdf/10.1007/979-8-8688-2183-7.pdf>
74. [XLS] queryResult - FDA <https://www.fda.gov/media/186044/download>
75. iScience - Cell Press [https://www.cell.com/iscience/pdf/S2589-0042\(24\)01488-3.pdf](https://www.cell.com/iscience/pdf/S2589-0042(24)01488-3.pdf)
76. [PDF] Guidelines 05/2020 on consent under Regulation 2016/679 ... - EDPB https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf
77. Guidelines 05/2020 on consent under Regulation 2016/679 - EDPB https://www.edpb.europa.eu/our-work-tools/our-documents/guidelines/guidelines-052020-consent-under-regulation-2016679_en

78. [PDF] Artificial Intelligence in Science and Society: the ... - ResearchGate https://www.researchgate.net/publication/387984873_Artificial_Intelligence_in_Science_and_Society_the_Vision_of_USERN/fulltext/678612371afb4e11f5e53583/Artificial-Intelligence-in-Science-and-Society-the-Vision-of-USERN.pdf
79. Advances in Neuroergonomics and Cognitive Engineering <https://link.springer.com/content/pdf/10.1007/978-3-030-51041-1.pdf>
80. Transcending the brain: is there a cost to hacking the nervous system? <https://pmc.ncbi.nlm.nih.gov/articles/PMC7425343/>
81. Brain augmentation and neuroscience technologies - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC9538357/>
82. Developing collaborative platforms to advance neurotechnology and ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC7610607/>
83. The risks and challenges of neurotechnologies for human rights <https://unesdoc.unesco.org/ark:/48223/pf0000384185>
84. TechDispatch #1/2024 - Neurodata <https://www.edps.europa.eu/data-protection/our-work/publications/techdispatch/2024-06-03-techdispatch-12024-neurodata>
85. Advancements in brain-machine interfaces for application in ... - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC11198002/>
86. [PDF] CCPR/C/GC/36 - ohchr https://www.ohchr.org/Documents/HRBodies/CCPR/CCPR_C_GC_36.pdf
87. UN expert calls for regulation of neurotechnologies to protect right to ... <https://www.ohchr.org/en/press-releases/2025/03/un-expert-calls-regulation-neurotechnologies-protect-right-privacy>
88. On Neurorights - PMC <https://pmc.ncbi.nlm.nih.gov/articles/PMC8498568/>
89. Ethics of Artificial Intelligence - AI | UNESCO <https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>
90. Recommendation on the Ethics of Artificial Intelligence <https://unesdoc.unesco.org/ark:/48223/pf0000386510>
91. Wave of State Legislation Targets Mental Privacy and Neural Data <https://www.cooley.com/news/insight/2025/2025-05-13-wave-of-state-legislation-targets-mental-privacy-and-neural-data>
92. AI Act Service Desk - Article 5: Prohibited AI practices <https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-5>
93. The EU Commission Guidelines on prohibited AI practices under the ... <https://www.lexology.com/library/detail.aspx?g=698e4e68-aaea-4a7a-bea6-a3748e277dc8>

94. [PDF] AI Act: Prohibited AI Practices Become Applicable - EY <https://www.ey.com/content/dam/ey-unified-site/ey-com/en-gr/technical/tax/tax-alerts/2025/02/ai-act-en-v3.pdf>
95. Commission publishes the Guidelines on prohibited artificial ... <https://digital-strategy.ec.europa.eu/en/library/commission-publishes-guidelines-prohibited-artificial-intelligence-ai-practices-defined-ai-act>
96. The European Commission's Guidance on Prohibited AI Practices <https://www.lexology.com/library/detail.aspx?g=077adba5-d5a4-4931-ba56-fce2fccbd236>
97. What Is "Thought"? Interpreting and Constructing Article 18 ICCPR ... https://link.springer.com/chapter/10.1007/978-3-031-91466-9_8
98. (PDF) Freedom of thought in Europe: do advances in 'brain-reading ... https://www.researchgate.net/publication/344098575_Freedom_of_thought_in_Europe_do_advances_in_'brain-reading'_technology_call_for_revision
99. UNESCO AI Ethics Recommendation 2021 | Artificial Intelligence <https://www.scribd.com/document/925184832/Ethics-of-Artificial-Intelligence-UNESCO>
100. Recommendation on the Ethics of Artificial Intelligence - Legal Affairs <https://www.unesco.org/en/legal-affairs/recommendation-ethics-artificial-intelligence>
101. Annual patent case law review - Lexology <https://www.lexology.com/library/detail.aspx?g=dcd846b5-af92-4afe-8190-811817adc9b6>
102. [XLS] queryResult - FDA <https://www.fda.gov/media/185288/download>
103. Forensic Brain-Reading and Mental Privacy in European Human ... <https://link.springer.com/article/10.1007/s12152-020-09438-4>
104. Human rights law: an incomplete but flexible framework to protect ... <https://www.tandfonline.com/doi/abs/10.1080/17579961.2024.2313796>
105. Recent Copyright Registrations Summary | PDF - Scribd <https://www.scribd.com/document/788821294/E-Register-December-2023>
106. European Commission Publishes the Guidelines on Prohibited ... <https://www.linkedin.com/pulse/european-commission-publishes-guidelines-prohibited-artificial-intelligence-iqs7f>
107. [PDF] CCPR/C/GC/36 - Official Document System <https://documents.un.org/doc/undoc/gen/g19/261/15/pdf/g1926115.pdf>
108. cmnt_vocab.txt - CMU School of Computer Science https://www.cs.cmu.edu/~ark/blog-data/data/blog_data_v1_0/dk/hbc_data/data/cmnt_vocab.txt
109. (PDF) Report of the Medical Image De-Identification (MIDI) Task Group https://www.researchgate.net/publication/369380004_Report_of_the_Medical_Image_De-Identification_MIDI_Task_Group_-_Best_Practices_and_Recommendations

110. 10.1007@978 3 030 14070 0 | PDF | Artificial Intelligence - Scribd <https://www.scribd.com/document/508734026/10-1007-978-3-030-14070-0>
111. Advanced Information Networking and Applications - Springer Nature <https://link.springer.com/content/pdf/10.1007/978-3-031-87784-1.pdf>
112. Appl. Sci., Volume 16, Issue 6 (March-2 2026) – 484 articles - MDPI <https://www.mdpi.com/2076-3417/16/6>
113. Understanding the Brain Function and Emotions - Springer Nature <https://link.springer.com/content/pdf/10.1007/978-3-030-19591-5.pdf>
114. A Survey on Large Language Model-based Agents in Autonomous ... <https://arxiv.org/html/2505.12786v2>
115. [PDF] Flexible Automation and Intelligent Manufacturing: The Human-Data https://www.researchgate.net/publication/364513721_A_Framework_for_Manufacturing_System_Reconfiguration_Based_on_Artificial_Intelligence_and_Digital_Twin/fulltext/6361714a431b1f530060007b/A-Framework-for-Manufacturing-System-Reconfiguration-Based-on-Artificial-Intelligence-and-Digital-Twin.pdf
116. [PDF] EDPB Document on response to the request from the European ... https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_replyec_questionnaireresearch_final.pdf
117. NLP Real Or Fake Text with 95% Acc - Kaggle <https://www.kaggle.com/code/jillanisofttech/nlp-real-or-fake-text-with-95-acc>
118. Capstone-Assignment - RPubs <https://rpubs.com/BushraT/1075426>
119. words-333333 - cs.Princeton <https://www.cs.princeton.edu/courses/archive/spring18/cos226/assignments/autocomplete/testing/words-333333.txt>
120. Dictionary of Acronyms and Technical Abbreviations - Academia.edu https://www.academia.edu/64999604/Dictionary_of_Acronyms_and_Technical_Abbreviations
121. (PDF) Neurotechnologies in the AI Act: Moving away from the ... https://www.researchgate.net/publication/399212126_Neurotechnologies_in_the_AI_Act_Moving_away_from_the_Neurorights_Debate